Insights into Imaging

Anatomiadi: A national, image-based anatomy competition for residents in Italy to enhance postgraduate education through interactive learning.

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Full Title:	Anatomiadi: A national, image-based anatomy competition for residents in Italy to enhance postgraduate education through interactive learning.
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Critical Relevance Statement Enter a one-sentence statement (max. 40 words) summarizing the key message of your article. It should explain how the article critically assesses an issue, and how it helps advancing the practice of clinical radiology.	Anatomiadi offers a scalable and reproducible blueprint for modernizing pedagogy in radiology. By integrating high-fidelity imaging with interactive game mechanics not only improves anatomical learning but also builds the foundation for a more collaborative and technologically adept generation of radiologists.

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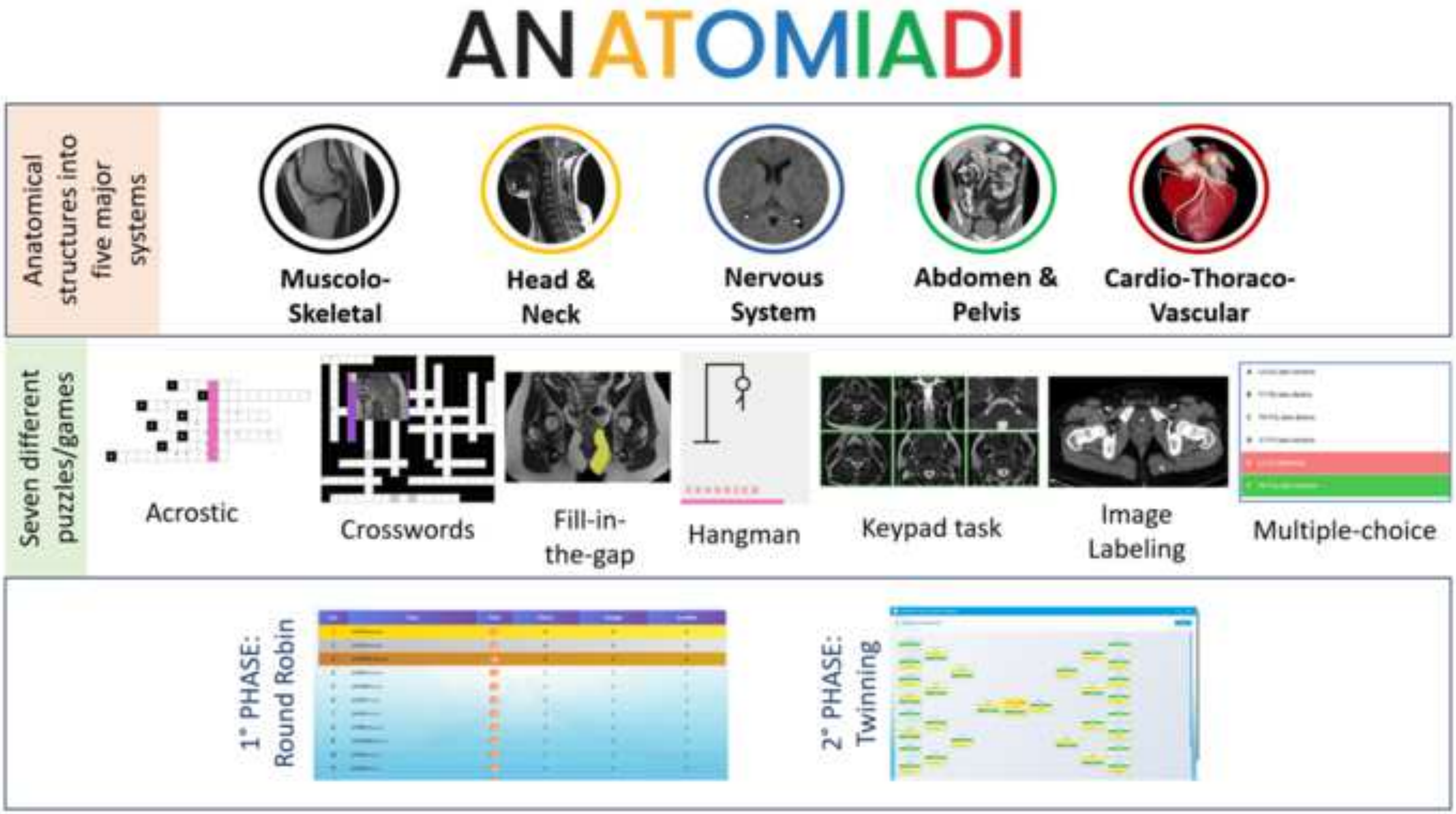
Graphical Abstract



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Dear Editor,

On behalf of the authors, I here submit the Original Manuscript entitled "**Anatomiadi: A national, image-based anatomy competition for residents in Italy to enhance postgraduate education through interactive learning**".

The authors of the present manuscript conceived, analysed and interpreted the data and gave their final approval of the version to be published. The authors had complete freedom to direct the analysis, without influence or censorship from any sponsor. The authors of this manuscript had no potential conflicts of interest and/or financial support.

We also warrant that this work represents original material and that no part of the work has been published or will be submitted for publication elsewhere unless and until a final decision is received by Insights Into Imaging.

Yours faithfully,

Trento, 27th april 2026

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ABSTRACT

Background - For radiology residents, the steep learning curve of cross-sectional anatomy (especially in CT and MRI) demands more efficient and engaging educational tools. In recent years, game-based learning (GBL) and gamification have emerged as powerful strategies to enhance postgraduate education by leveraging 'active recall' and 'spaced repetition' within a competitive framework. Here we describe the context of Anatomiadi and assess the perceived educational impact.

Methods - Image datasets were anonymized, archived and manually labeled to highlight specific anatomical structures. Anatomical structures were categorized into five major systems: Nervous, Head and Neck, Cardio-Thoraco-Vascular, Abdomino-Pelvic, and Musculoskeletal. Over 400 unique games were developed, categorized by anatomical system and tiered into three levels of difficulty: basic, intermediate, and advanced, including acrostic puzzles and crosswords. The specific digital platform developed was built as a web application. A structured feedback survey was designed and distributed to the participants after the event.

Results - Anatomiadi challenge registered an attendance of 141 participants representing 28 residency programs across Italy. The feedback survey indicated a generally positive perception across most items related to technical interaction and competition dynamics, the impact on learning image-based anatomy, satisfaction and engagement, peer recommendation and educational advocacy. The SWOT analysis highlighted specific weaknesses and threats.

Conclusion By integrating high-fidelity imaging with interactive game mechanics, Anatomiadi challenge not only fostered anatomical learning but also built the foundation for a more collaborative and technologically adept generation of radiologists.

Take Home Messages:

1. Anatomiadi is the first nationwide competition dedicated exclusively to radiological anatomy. It redefines the traditional competitive paradigm by shifting from an exclusionary model to an inclusive 'twinning' framework among residency programs.

2. The feedback survey results underscore that Anatomiadi successfully bridges the gap between theoretical study and practical engagement, transforming a traditionally dense subject like anatomy into a dynamic, social, and highly motivating professional learning experience.

Keywords

Diagnostic imaging; Education; Learning; Post-graduate teaching; Gamification

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INTRODUCTION

Anatomy represents the fundamental cornerstone of diagnostic imaging and clinical practice in radiology. Traditional teaching models, often based on passive memorization of 2D atlases, frequently fail to bridge the gap between theoretical knowledge and the complex 3D interpretative skills required in a modern clinical setting [1]. The mastery of radiological anatomy is, indeed, no longer a static exercise in memorization but a dynamic process of spatial synthesis. In the era of high-resolution multidetector CT and advanced MRI, the volume of anatomical data that a resident must process is unprecedented. Traditional pedagogical models often struggle to keep pace with this technological evolution, leading to a gap between theoretical knowledge and clinical diagnostic performance [2-4].

For radiology residents, the steep learning curve of cross-sectional anatomy (especially in CT and MRI) demands more efficient and engaging educational tools. In recent years, game-based learning (GBL) and gamification have emerged as powerful strategies to enhance postgraduate education by leveraging 'active recall' and 'spaced repetition' within a competitive framework [5-8]. By integrating competitive elements and interactive feedback, these methods can significantly increase trainee engagement, enhance long-term retention, and accelerate the development of visual pattern recognition.

Within this framework, we developed the idea of a context using image-based puzzle games to challenge different radiology residency schools in Italy on identification of anatomical structures and knowledge of anatomical rules. As such we coined the term “Anatomiadi” to recall the spirit of Olympic games and refer to a specific competitive yet collaborative environment aiming at promoting a faster, more intuitive mastery of radiological anatomy. The rationale behind this program is that the synergy between educational psychology and advanced visualization technology not only streamlines the learning curve in postgraduate students but also fosters a standardized, high-level competency across residency schools, ultimately improving diagnostic confidence in future radiologists.

The objective of this study is to describe and explain the context of Anatomiadi and assess the educational impact and perceived effectiveness of the first edition of the competition by analyzing the results of the post-event feedback survey. Specifically, we aim to assess the perspectives of participants including residents, expert radiologists playing as referees in the games and academic staff, regarding the impact of Anatomiadi on knowledge, its pedagogical structure, and the role of integrating gamified technology in postgraduate radiology training.

METHODS

Image Database and Tagging

The main asset of Anatomiadi centers on a high-fidelity database comprising high-resolution Computed Tomography (CT), Magnetic Resonance Imaging (MRI), Ultrasound (US), and conventional X-ray images. These 2D and 3D datasets were anonymized, archived and manually labeled by residents of the program of the University of Trento (Figure 1) to highlight specific anatomical structures, then validated by three senior radiologists (C.C.Q., M.N., A.F.), each with over 20 years of experience. The step of validation ensured pedagogical relevance, specifically for complex anatomical structures and variants.

Images were optimized for rapid web-based visualization while maintaining full diagnostic quality. Each entry was indexed in a structured table, linking the image code to up to three descriptors (e.g., 'right,' 'hepatic,' 'lobe' for an axial abdominal CT) (Figure 1). Descriptive adjectives (e.g., 'superior', 'anterior') were included to refine identification. This methodology resulted in the segmentation of over 1,800 anatomical structures, associated with more than 4,300 descriptors. Structures were categorized into five major systems: (1) Nervous, (2) Head and Neck, (3) Cardio-Thoraco-Vascular, (4) Abdomino-Pelvic, and (5) Musculoskeletal. Overlapping classifications were permitted; for instance, the right meningeal artery at the foramen spinosum was assigned to the Head and Neck, Cardio-Thoraco-Vascular, and Nervous categories. Finally, labeling conventions (filled vs. empty areas) were employed to distinguish between a structure's content (e.g., a nerve) and its container (e.g., a canal), preventing ambiguity.

Puzzle Development and Gamification

To transform traditional anatomical identification into a high-engagement educational challenge, we leveraged the interactive tools provided by the online platform Puzzel.org, founded by Daan Weustenraad (https://puzzel.org). This allowed for the development of diverse enigmatics and puzzles, including customized crosswords, drag-and-drop labeling, and visual recognition tasks to evaluate not only precision but also response speed of residents by including tasks to be performed under time constraints. Ultimately, seven game formats were identified as most effective for anatomical instruction: acrostic puzzles, crosswords (Figure 2), fill-in-the-gap, hangman, keypad tasks, image labeling, and the standard multiple-choice quizzes. Over 400 unique games were developed, categorized by anatomical system and tiered into three levels of difficulty: basic, intermediate, and advanced.

The competition was structured as a head-to-head challenge between two teams, each composed of a maximum of four players (residents). Participants engaged with the same puzzle, with turns alternating after every response regardless of accuracy. The sessions were conducted via interactive touchscreen interfaces **(**Samsung Flip Pro Digital Flipboard 65’’), which facilitated real-time collaboration and engagement.

Faculty members served as moderators, responsible for scoring and ensuring adherence to the specific time constraints established for each game. Additionally, a dedicated staff member was present to provide technical support and manage logistical requirements throughout the exercise.

Competitive Platform and Scoring System

The competition was run into two phases: a first round-robin phase on day 1 when teams competed in a league format to establish initial rankings and a second tournament phase when, by a progressive twinning mechanism, losing teams merged with the winners, forming increasingly larger macro-teams. This collaborative-competitive structure culminated in a Grand Finale between two opposing macro-coalitions, fostering a sense of national community among the participants while maintaining the intensity of the competition.

We developed a dedicated digital infrastructure to manage the real-time progression of the two phases. The platform featured a dynamic dashboard for the live display of progressive results and scoring.

The digital platform developed for the Anatomiadi was built as a web application, accessible via the dedicated domain www.anatomiadi.eu (Figure 3). The architecture follows a three-tier model: the front-end was developed using Angular, optimized for large screens while remaining responsive for mobile devices; the back-end was based on Node.js with the Express framework; and data persistence was handled by PostgreSQL. To ensure real-time updates during the context a critical requirement for managing competitive sessions, a WebSocket-based communication layer was integrated. The entire application was containerized using Docker and orchestrated with Docker Compose, adopting a Git-based automated deployment workflow. The infrastructure was hosted on a virtual machine running Ubuntu 22.04 LTS, configured with 4 vCPUs, 8 GB of RAM, and 40 GB of SSD storage, sized to support an estimated load of 50 to 100 concurrent users. Access to the platform was managed through internal authentication, with separate credentials for administrators and referees.

Feedback Survey

As a follow-up of the event, a feedback survey was designed and distributed to the participants of the Anatomiadi including residents, referees/experts and staff members. The survey aimed to capture data on different domains regarding the educational value, gamification and engagement, platform usability and team collaboration, technical workflow, logistics, scientific accuracy, fairness and quality assessment. For most questions a 5-point Likert scale was used to categorize responses in the different domains.

The survey comprised 30 questions structured into four sections:

Section 1 - Technical Interaction and Competition Dynamics. Q1, Was the tactile responsiveness of the touchscreen interfaces adequate for the specific requirements of the gamified tasks?; Q2, Was the visual layout and User Interface (UI) formatting optimized for large-scale monitor display and participant interaction?; Q3, Did the radiological images (CT, MRI, US) maintain sufficient spatial resolution and diagnostic quality to meet the educational objectives?; Q4, Were the highlighted anatomical structures clearly discernible within the provided radiological datasets?; Q5: Was the distribution of anatomical regions (e.g., nervous, abdomino-pelvic, musculoskeletal) appropriately balanced across the competitive rounds?; Q6, Was the progressive increase in question difficulty from Day 1 to Day 2 perceived as appropriate and pedagogically effective?; Q7, Were the time limits allocated for each task sufficient to allow for accurate diagnostic reasoning while maintaining competitive pressure?; Q8, Was the presence of a dedicated proctor/assistant at each station perceived as helpful for technical support and procedural guidance?; Q9, Is the presence of a dedicated assistant considered essential for the successful execution of the game-based tasks?

Section 2 – Impact on Learning Radiological Anatomy. Q10, What was the level of knowledge acquired at the end of the event?; Q11, What was the level of knowledge required to solve puzzles?; Q12, How do you rate the overall contribution to learning radiological anatomy?; How do you rate the specific contribution to learning anatomy of the acrostic puzzles (Q13), crosswords (Q14), image labeling (Q15), fill-in-the-gap (Q16), multiple-choice quizzes (Q17), hangman (Q18), keypad (Q19)?

Section 3 – Perceived Satisfaction and Engagement. How do you rate the level of satisfaction/engagement of the acrostic puzzles (Q20), crosswords (Q21), image labeling (Q22), fill-in-the-gap (Q23), multiple-choice quizzes (Q24), hangman (Q25), keypad (Q26)?

Section 4 – Peer Recommendation and Educational Advocacy. Q27, open-ended question for general comments about Anatomiadi; Q28, open-ended question to comment on strengths, weaknesses, opportunities and threats aimed to carry out a SWOT analysis; Q29, Would you recommend a resident in radiology to participate to Anatomiadi?; Q30, which year of residency is the best target for Anatomiadi in the training curriculum (first to fourth year)?.

Statistical analysis

Descriptive statistics were calculated for all survey items assessed with the 1-5 Likert scale. Composite scores were generated for thematic aggregate analysis of survey items (Q1-Q4, Q5, Q6-Q9, Q10-Q12, Q13-Q19 and Q20-Q26). To compare the mean scores across the thematic clusters, a One-Way Analysis of Variance (ANOVA) was conducted. Likert scale data were treated as interval data for the purpose of this analysis. The assumption of homogeneity of variances was assessed using Levene's Test, and post-hoc comparisons were performed using Tukey's HSD (Honestly Significant Difference) to identify specific pairwise differences between groups. Statistical significance was set at $p<0.05$.

RESULTS

Anatomiadi was held in Rovereto (Trento, Italy) on December 11-12 2025, with an attendance of 141 participants representing 28 residency programs across Italy. The cohort included 98 residents, supported by 29 referees and 14 staff members. A total of 63 (44.7%) participants completed the feedback survey within one month after the event. Respondents represented, proportionally balanced, the three different groups, with the majority being residents (74.6%, 47/63), followed by referees (15.9%, 10/63) and staff members (9.5%, 6/63). Results are summarized in table 1.

The survey results indicated a generally positive perception across most items. The highest level of agreement was observed for Q9, where 92.1% of participants reported positive responses (Mean ±

SD = 4.48 ± 0.88). Similarly, Q3 and Q6 showed strong consensus with 87.3% and 84.1% (Mean ± SD = 4.24 ± 0.93) agreement, respectively.

Conversely, Q2 was the most divisive item, yielding the lowest agreement rate (47.6%), the lowest mean score (Mean ± SD = 3.27 ± 1.11) and the highest frequency of neutral responses (17.3%). Items Q5 and Q7 also showed higher levels of disagreement compared to the rest of the respondents, with approximately 28% and 24% of respondents expressing negative views.

Notably, Q5 and Q7 exhibited the highest variance (1.18 and 1.14 respectively), suggesting a lack of consensus among the sample.

The One-Way ANOVA revealed a significant effect of thematic grouping on participant responses ($p<0.001$). Post-hoc comparisons using the Tukey HSD test indicated that the mean score for Q6-Q9 (Mean ± SD = 4.07 ± 1.00) was significantly higher than Q1-Q4 (Mean ± SD = 3.71 ± 1.01, $p=0.01$) and Q5 (Mean ± SD = 3.37 ± 1.18, $p<0.001$). No statistically significant difference was found between Q1-Q4 and Q5, suggesting that assistance of the dedicated staff was crucial for the perceived success of interaction with touchscreens.

The descriptive analysis of items Q10 through Q19 revealed a consistently positive trend, with all mean scores exceeding the theoretical midpoint of 3.0. The highest levels of agreement (51/63 positive responses) were recorded for Q11 (Mean ± SD = 4.10 ± 0.90) and Q14 (Mean ± SD = 4.06 ± 0.86). Q10 exhibited the lowest mean score (Mean ± SD = 3.63 ± 0.99). Q10 and Q13 received the highest number of neutral responses (20 out of 63, 31.7%). Q17 received the highest number of negative responses (9 out of 63, 14.3%).

Analysis of items Q20-Q26 revealed that Q21 reached the highest level of consensus (Mean ± SD = 4.48 ± 0.69) with 88.9% of respondents reporting positive scores on crosswords. Q23 showed fill-in-the-gap as the least favoured puzzle (Mean ± SD = 3.30 ± 1.19), showing an agreement rate of only 50.8% and the highest number of disagreements (n = 15). Q26 (keypad) exhibited the highest variance (Mean ± SD = 3.68 ± 1.28), suggesting a conflictual perception among participants.

Open-ended feedback (Q27) revealed an overwhelmingly positive reception of the Anatomiadi competition. The comments could be synthesized into four major themes: 1. innovation and game-based format as a powerful stimulus to identify individual knowledge gaps and to drive a rigorous study of anatomy; 2. networking at national level as the competition provided a framework to connect and compare different levels of expertise; 3. engagement and motivation as many comments praised the value of “healthy competition” and “team building”; 4. inclusivity of the event as it is uncommon to provide such an experience in post-graduate training. While some technical touch-screen issues and procedural ambiguities were noted, they did not diminish the overall “extremely positive judgment" of the initiative.

Responses to Q28 allowed to build a SWOT analysis presented in Table 2.

The evaluation of the event's perceived value yielded a unanimous positive response; indeed 100% of the participants recommended that the competition being repeated in future academic years (Q29). Responses to Q30 showed that, based on the required level of knowledge, the event would be recommended to residents in their 3rd (54 out of 63 responses) and 4th year (48 out of 63 responses) of post-graduate education. In fact, only 7 respondents recommended the event for 1st-year residents. This distribution also suggested that the competition's complexity and collaborative elements can be particularly well-received by those in more advanced stages of training.

DISCUSSION

The results of the feedback survey on the Anatomiadi competition demonstrate that gamification is a powerful pedagogical tool for radiological anatomy, significantly enhancing engagement and peer-to-peer knowledge transfer among residents [9]. The high level of satisfaction and the unanimous recommendation for future editions (100% agreement among respondents) underscore the demand for innovative, interactive formats in post-graduate medical education. In fact, while gamification of anatomy has demonstrated to increase learner engagement, motivation, and knowledge retention in medical graduate training [10, 11], its role in postgraduate radiology training has been frequently undervalued; and a high level of anatomical proficiency is a critical determinant of diagnostic accuracy and clinical decision-making.

A key finding of this study was the critical role of human-mediated technical support. While participants reported high satisfaction with the digital interfaces and image quality (Q3, Q9), the significant divergence in opinions regarding the large-scale UI (Q2) suggests that the physical layout of the competition workstations required manual intervention from proctors to ensure a smooth experience. The presence of dedicated assistants (Q6–Q9) was perceived as more effective feature than the standalone technical interaction (Q1–Q4), highlighting that in high-pressure gamified environments, technical reliability must be paired with immediate human troubleshooting. In addition these results support the idea that efforts to adapt the environment to learners' specific needs can increase motivation and engagement [12].

Pedagogically, the "twinning" mechanism transformed a standard competition into a collaborative learning ecosystem. By merging teams, the event facilitated "inter-institutional benchmarking," allowing residents from 28 different programs to compare expertise and bridge knowledge gaps through active recall. The preference for crosswords (Q14, Q21) over more traditional formats like multiple-choice quizzes (Q17) or fill-in-the-gap tasks (Q23) suggests that puzzles requiring constructive retrieval of anatomical terms are more effective for long-term retention than recognition-based tasks.

Furthermore, the data regarding the target audience (Q30) indicates that the competition's complexity is best suited for senior residents (3rd and 4th years). This aligns with the idea that game tasks require a pre-existing knowledge and experience in image reading and interpreting skills.

Despite the positive outcomes, several limitations must be acknowledged. First, the sample size of survey respondents (n=63) represents only 44.6% of total participants, which may introduce a self-selection bias toward those who had a more positive experience. Second, the technical disparities noted in touchscreen responsiveness and the lack of a standardized synonym-recognition engine for anatomical terms (e.g., Latin vs. Italian nomenclature) may have introduced "noise" into the scoring system, potentially affecting the perceived fairness of certain rounds. Third, the cross-sectional nature of the survey measures "perceived" learning and satisfaction rather than objective long-term knowledge retention. While participants felt they acquired significant knowledge (Q10), a follow-up assessment would be required to quantify the actual educational impact. Finally, the logistical challenges of managing large "macro-teams" in the final rounds may have diluted individual participation, suggesting that future iterations should refine the collaborative workflow to ensure equitable engagement for all team members.

The Anatomiadi competition may represent a paradigm shift in postgraduate radiological education, successfully bridging the gap between theoretical anatomical knowledge and practical diagnostic proficiency through gamification. The post-event analysis demonstrates that a multi-institutional, competitive-collaborative framework significantly enhances learner engagement, accelerates the mastery of complex 3D structures, and fosters a robust national professional network among residents. While the pilot edition highlighted areas for technical refinement - specifically regarding UI/UX optimization, automated synonym recognition, and the standardization of refereeing protocols - the overwhelming peer-to-peer recommendation rate validates the model's effectiveness. The unique 'progressive twinning' mechanism proved to be a powerful tool for social learning, transforming individual competition into a collective educational experience. The 100% recommendation rate was particularly notable given the diverse institutional backgrounds of the residents. This suggests that the benefits of the competition - specifically active recall and inter-institutional benchmarking -transcend local curricular differences, making it a universally applicable educational tool.

In conclusion, the Anatomiadi format offers a scalable and reproducible blueprint for modernizing pedagogy in radiology. By integrating high-fidelity imaging with interactive game mechanics not only improves anatomical learning but also builds the foundation for a more collaborative and technologically adept generation of radiologists. Future iterations, incorporating the feedback-driven improvements identified in this study and eventually empowerment through artificial intelligence [13], have the potential to set a new gold standard for radiological training across Europe.

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Table 1. Survey results presented in terms of agreement (positive). Disagree = Likert Scale 1 and 2; Neutral = Likert Scale 3; Agree = Likert Scale 4 and 5. SD = Standard deviation.

Questions	Disagree	Neutral	Agree	% Positive	Mean	SD
Q1	11	12	40	63.5	3.49	1.01
Q2	16	17	30	47.6	3.27	1.11
Q3	5	3	55	87.3	4.19	1.01
Q4	4	11	48	76.2	3.90	0.91
Q5	18	10	35	55.6	3.37	1.18
Q6	2	8	53	84.1	4.24	0,93
Q7	15	13	35	55.6	3.41	1.14
Q8	6	9	48	76.2	4.13	1.07
Q9	3	2	58	92.1	4.48	0.88
Q10	6	20	37	58.7	3.63	0.99
Q11	4	8	51	81.0	4.10	0.90
Q12	8	14	41	65.1	3.73	1.10
Q13	5	20	38	60.3	3.67	0.97
Q14	3	9	51	81.0	4.06	0.86
Q15	5	15	43	68.3	3.84	1.02
Q16	5	14	44	69.8	3.95	1.01
Q17	9	11	43	68.3	3.67	1.09
Q18	7	13	43	68.3	3.78	0.96
Q19	8	12	43	68.3	3.86	1.11
Q20	5	11	47	74.6	3.92	0.94
Q21	1	6	56	88.9	4.48	0.69
Q22	5	16	42	66.7	3.81	0.98
Q23	15	16	32	50.8	3.30	1.19
Q24	11	12	40	63.5	3.65	1.18
Q25	10	7	46	73.0	3.87	1.22
Q26	13	12	38	60.3	3.68	1.28

Table 2. SWOT Analysis summarizing responses to open-ended Q28.

STRENGTHS	WEAKNESSES
• Pedagogical Innovation: Successful use of active recall and "diagnostic" learning over passive study. • Collaborative "Twinning": Unique hybrid model that fosters peer-to-peer teaching across different schools. • High Engagement: Gamification effectively bridges the gap between abstract textbook theory and clinical practice. • Pre-event Preparation: The setup phase is itself a high-value learning tool for residents.	• Technical Latency: Touchscreen lags and software bugs affect the high-pressure user experience. • Rule Inconsistency: Subjective officiating regarding synonyms and time management creates "friction." • UI/UX Ergonomics: Screen layouts and visibility favoured certain positions at the table. • Scoring Imbalance: Points-to-difficulty ratio issues where specific games create "insurmountable" leads.
OPPORTUNITIES	**THREATS**
• Automation: Implementing an automated synonym-recognition engine to standardize scoring. • Infrastructure Scaling: Developing a proprietary, unified OS to replace unstable third-party components. • Curriculum Expansion: Integrating topographical anatomy and diverse subspecialties to broaden the scope. • Post-Game Learning: Adding immediate feedback loops (visualizing correct answers) to reinforce knowledge retention.	• Perceived Inequity: If officiating and scoring aren't standardized, participants may lose trust in the "fairness" of the competition. • Competitive Fatigue: Insurmountable leads mid-game can lead to participant disengagement. • Technological Risk: Hardware or software failure during high-stakes rounds could damage the event's professional reputation.

Figures Legend

Figure 1. Example images with colored overlaid areas to label the hepatic caudate lobe on an axial CT image (a) and the capitate bone on a wrist conventional X-ray (b).

Figure 2. Layout of the Crosswords game. A maximum time of 600 seconds was given to complete it. Single response time was 10 seconds for basic level and 15 seconds for intermediate and advanced level of difficulty.

Figure 3. Live display of the league to show the real-time progression during the second day of the Anatomiadi competition.

Figure 1.

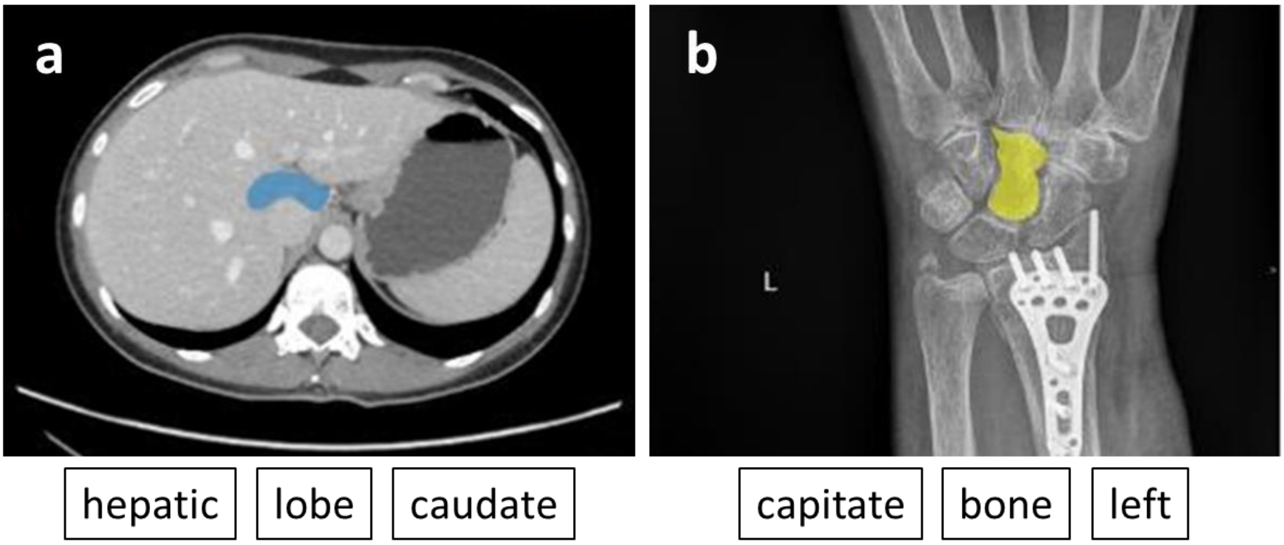

Figure 2.

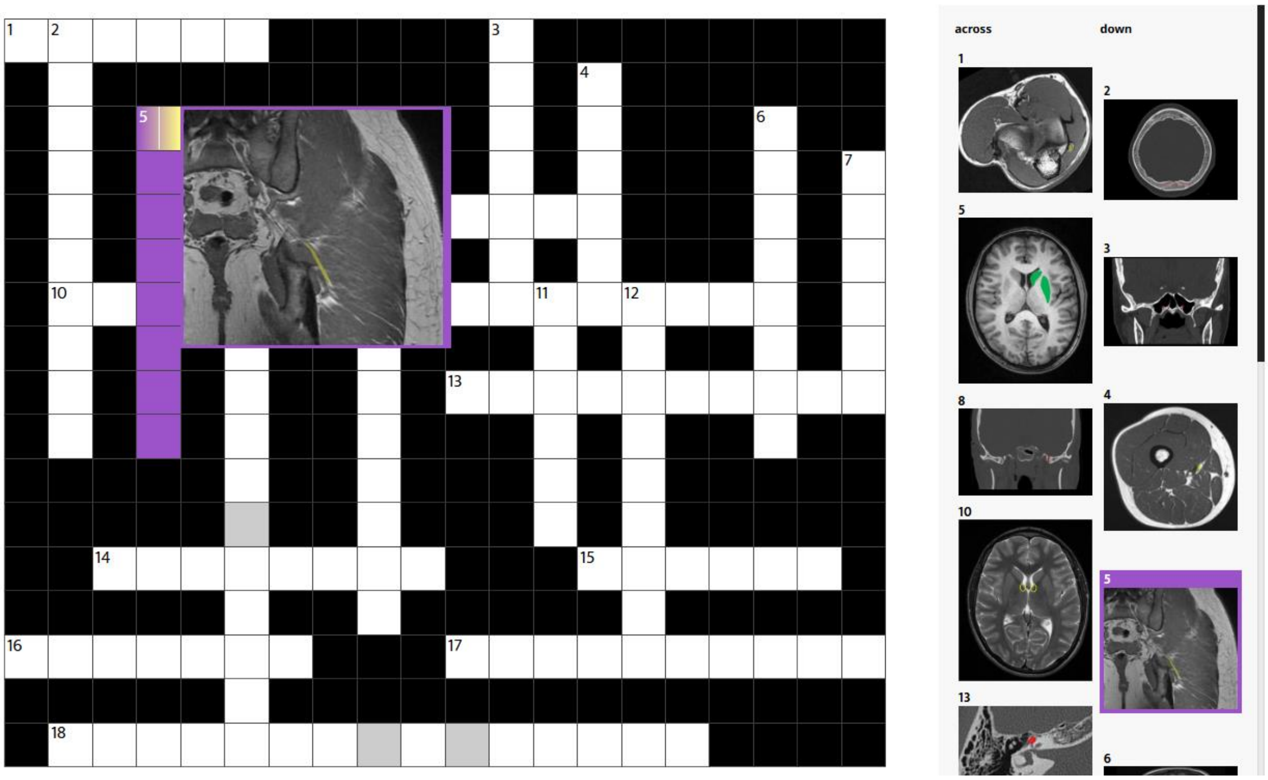

Figure 3.


ANATOMIADI: I giochi di Anatomia Radiologica
HOME
ADMIN
Tabellone Anatomiadi
Aggiorna
SEMIFINALE
UniSR
vs
UniTS
FINALE
UniTS
vs
UniPA
SEMIFINALE
UniPA
vs
UniGE

Declarations:

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable

Availability of data and material

Data used for analysis can be made available to interested parts upon request.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

CCQ developed the project idea, managed the database, built the puzzles and games format, developed the feedback survey, conducted statistical analysis and wrote the first draft of the manuscript; MN and AF collaborated in the project development, validated anatomical structures, built puzzles and games and tested the puzzles gaming; MA, FDR, MF, PG, GG, DR, FS and DV collected images, populated the database, labelled anatomical structures and tested the puzzle gaming;

AA, AC, VD and PG developed the digital infrastructure and the scoring system and collaborated to the first draft of the manuscript. All authors revised the manuscript and approved the final version.

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